



Experimental Liquidus Studies of the Binary Pb-Cu-O and Ternary Pb-Cu-Si-O Systems in Equilibrium with Metallic Pb-Cu Alloys

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Abstract Phase equilibria of the Pb-Cu-Si-O system have been investigated in the range 933–1938 K (660–1665 °C) for oxide liquid (slag) in equilibrium with solid Cu metal, liquid Pb-Cu alloy, or both solid and liquid metals, and solid oxide phases: (a) quartz, tridymite, cristobalite (SiO₂); (b) cuprite (Cu₂O); (c) lead silicates (PbSiO₃, Pb₂SiO₄, Pb₁₁Si₃O₁₇); (d) lead oxide (massicot, PbO); and (e) copper plumbite (Cu₂PbO₂). High-temperature equilibration on silica or copper substrates, followed by quenching and direct measurement of Pb, Cu and Si concentrations in the liquid and solid phases with the electron probe x-ray microanalysis (EPMA) has been used to accurately characterize the system in equilibrium with Cu or Pb-Cu metal. All results are projected onto the PbO–“CuO_{0.5}”–SiO₂ plane for presentation purposes. The present study is a continuation of the previous investigation of this system by the authors in a part of the silica and cuprite primary phase fields. Present data were later used to develop the thermodynamic models for all phases in this system.

Keywords copper · lead · liquidus · PbO–CuO_{0.5}–SiO₂ · phase diagrams · silica · slags

1 Introduction

The present study is a continuation of the investigation of the PbO–“Cu₂O”–SiO₂ system in equilibrium with metal by Shevchenko, Nicol et al.^[1] This subsystem represents an important part of the overall integrated thermodynamic modelling and experimental research program for the multicomponent Pb–Zn–Fe–Cu–Si–Ca–Al–Mg–O system. In the present study, the range of compositions investigated is significantly expanded towards high-temperature area (up to two immiscible liquids over cristobalite at 1665 °C) and high-PbO area (massicot, lead silicates, and copper plumbite primary phase fields), as well as the “Cu₂O”–PbO binary, which is important for future thermodynamic modelling. The literature on the PbO–“Cu₂O”–SiO₂ system has been reviewed in Ref 1.

2 Experimental Technique and Procedure

2.1 Experimental Procedure and Samples Examination

The experimental procedure and apparatus have been described in detail in previous publications by PYROSEARCH.^[2–4] The initial mixtures were made by mixing high-purity powders Cu₂O (99.9 wt.% purity), PbO (99.9 wt.% purity), SiO₂ (99.9 wt.% purity), supplied by Alfa Aesar, MA, USA or Heysham, Great Britain. Cu powder (99.9 wt.% purity; supplied by Alfa Aesar, MA, USA) or Pb powder (99.95 wt.% purity; supplied by Sigma Aldrich, MO, USA) were added with 10–20 wt.% in excess of the mass of the oxide mixture to ensure the metallic phase was present at the final equilibrium. Less than 0.5 g of mixture was used in each equilibration experiment. The

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initial compositions of the mixtures were selected so that one or more crystalline phases would be present in equilibrium with liquid slag. The volume fraction of solids was targeted to be below 50%, and preferably about 10%, to achieve rapid equilibration and useful outcomes from quenching (it was found that the solids serve as heterogeneous nucleation centers, and the minimum distance between them should exceed 10–20 micron to ensure that amorphous slag of uniform composition, unaffected by dendritic crystal growth, is obtained). The optimum fraction of solids was found to be dependent on the slag composition and the equilibrium temperature, these factors influence the slag viscosity, the kinetics of crystallization of the melt and equilibration time. An iterative procedure involving preliminary experiments was often needed to achieve the targeted proportions of the phases for a given final temperature and composition, since the exact liquidus temperatures at given compositions were not initially known. The mixtures were pressed into pellet using a tool steel die.

Three types of substrates were used for equilibration, depending on the conditions:

1. Silica ampoules (5 cm long, 1.3 cm outer diameter) sealed under vacuum (~ 0.001 atm) with the samples in the cristobalite/tridymite/quartz primary phase fields were used to prevent the loss of the volatile lead species from the system, as well as oxidation of the samples by residual oxygen in the gas atmosphere. Pelletizing of the powder mixture was still useful in this case, to avoid the presence of loose powder particles, which could be preferentially extracted from the ampoule during evacuation, and to minimize occupational health risk in the preparation of the samples.
2. Open solid copper support was used for samples in other primary phase fields, equilibrated at temperatures lower than 1000 °C, enabling direct contact of the slag with the quenching medium on cooling. The pellets were wrapped in 40 mesh copper gauze (net), which was then folded into an envelope shape (15 by 15 mm); the liquid samples were held in the support by surface tension. Oxidation of the alloy was prevented by the use of a flowing gas stream of high-purity Argon (99.999% purity; supplied by Coregas, NSW, Australia). This support type could not be used at temperatures higher than 1273 K (1000 °C) because of softening and melting of the copper metal in the support envelope.
3. Iridium (99.99%) wire substrate was used for the samples containing Pb–Cu liquid alloy at low temperatures in the low-Cu₂O range of compositions.

All equilibration experiments were carried out in a vertical reaction tube (impervious recrystallized alumina, 30-mm inner diameter) within an electrical resistance silicon carbide (SiC) heated furnace. The samples were placed immediately adjacent to a working thermocouple in a recrystallized alumina sheath in the uniform hot zone of the furnace to monitor the actual sample temperature. The working thermocouple was calibrated against a standard thermocouple (supplied by the National Measurement Institute of Australia, NSW, Australia). The overall absolute temperature accuracy of the experiment was estimated to be ± 3 K.

The samples were suspended in the hot zone of the furnace by Kanthal support wire (Fe–Cr–Al alloy, 0.7 or 1-mm diameter). For temperatures higher than 1773 K (1500 °C), a 20–25 cm Mo wire was used to extend the support, so that the Kanthal wire is not in of the hot zone. The samples were pre-melted for 30–60 min at 20–100 K above the equilibration temperature, to form a homogeneous liquid slag. This was followed by equilibration at the final target temperature for the required time. At the end of the equilibration process, the samples was released and rapidly quenched into CaCl₂ or NaCl-containing brine (< 253 K (-20 °C)). The samples were then washed thoroughly in water and ethanol before being dried on a hot plate, mounted in epoxy resin, and cross-sectioned using conventional metallographic polishing techniques.

The samples were examined by optical microscopy, carbon-coated, and the phase compositions were measured using an electron probe x-ray microanalysis technique with wavelength dispersive detectors (JEOL 8200L EPMA; Japan Electron Optics Ltd., Tokyo, Japan). The EPMA was operated with 15 kV accelerating voltage and 20 nA probe current using the standard Duncumb–Philibert atomic number, absorption, and fluorescence (ZAF) correction supplied by JEOL.^[5–7] Copper (Cu), cuprite (Cu₂O), quartz (SiO₂), wollastonite (CaSiO₃) (supplied by Charles M. Taylor Co., Stanford, CA), and K-456 lead silicate glass (71 wt. pct. PbO, supplied by NIST) standards were used for calibration of EPMA. The concentrations of metal cations were measured with EPMA; no information on the oxidation states of the metal cations was obtained. Copper oxide concentrations were recalculated as Cu₂O for presentation purposes; this oxidation state was selected according to the information on the slags in equilibrium with metal in the Cu–Si–O system.^[8] The molar ratios of the oxides were used in the graphical presentation of the data.

The ability to quench the liquid slag phase to ambient temperature without the onset of crystallization was found to depend on the composition of the slag and the equilibration temperature. The most significant problems were observed with low-silica slags in the cuprite primary phase field at high temperatures, at or above 1173 K (900 °C),

where the glassy homogeneous areas were only a small proportion of the melt, believed to be formed at the surfaces directly contacting the quenching medium. For these samples the approach taken to obtain accurate, repeatable, and objective measurements of the average compositions of the liquid slag phase by the use of EPMA was similar to that described by Nikolic et al.:^[9] an average of at least 20 points in the best quenched area (usually, near the surface) was used, additionally controlled by standard deviation of composition not exceeding 1 mol.%. In cases where melt crystallization was rapid, the experiments were repeated until an area of sufficiently well-quenched microstructure was found. Increasing the probe diameter to 5–20 micron resulted in reduced variability of the measured liquid slag compositions.

The standard ZAF correction was further improved following an approach similar to the one previously described.^[10–12] To check the accuracy of ZAF correction for the “CuO_{0.5}”-SiO₂ system, a reference material of fixed composition is desired in the intermediate range of compositions between pure oxides. Two types of materials can be used for that: (a) quenched slag (analyzed using alternative technique, such as wet chemistry); (b) stoichiometric compounds. There are significant problems in quenching of a homogeneous slag in the “CuO_{0.5}”-SiO₂ system as well as the adverse effect of possibility of variable copper oxidation states and substrate interaction on the accuracy of slag composition determined by wet chemistry. Moreover, there are no stable compounds known in “CuO_{0.5}”-SiO₂ or “CuO”-SiO₂ systems. However, it was found that in the Cu-Si-O-H system the mineral diopside has well-defined stoichiometry CuSiO₃·H₂O, and can be dehydrated to a metastable CuSiO₃ compound at 600–650 °C^[13–16] before subsequent slow decomposition into SiO₂ and CuO or Cu₂O + O₂. Single crystals of natural diopside (Altyn-Tyube diopside deposit, Altyn-Tyube area, Karaganda, Kazakhstan) were analyzed using EPMA both directly and after dehydration at 600–650 °C for 2 h (resulted in metastable CuSiO₃). No contamination has been detected, and the total wt.% (CuO + SiO₂) increased from 90% in the original diopside to 101% the dehydrated diopside (Fig. 1), close to the theoretical values (note that the excess total over 100% for dehydrated diopside and the 0.5% change in mol.% Si/(Cu + Si) on heating are also due to ZAF correction inaccuracy). The dehydrated diopside was taken as more reliable standard due to absence of water affecting the ZAF correction, and located most closely to the Cu₂O-SiO₂ system. Importantly, 51.8 mol.% SiO₂ was observed, which means overestimation of SiO₂ concentration by 1.8%. Due to absence of any more secondary reference points for compositions in the Cu₂O-SiO₂ system, it is reasonable to assume a simple parabolic dependence of the correction on mol.% SiO₂, with maximum correction

value for Si:Cu = 1:1 ratio (diopside) and gradually decreasing to zero in directions of both pure components. All new as well as previously published at PyroSearch data in the Cu₂O-SiO₂ binary, (PbO, ZnO, FeO_x, CaO, Al₂O₃, MgO,...)-Cu₂O-SiO₂ ternaries and multicomponent systems have to be corrected. The correction formula for the pseudobinary CuO_x-SiO₂ system is:

$$x_{\text{SiO}_2}^{\text{corrected}} = x_{\text{SiO}_2} - 0.018 \cdot 4 \cdot x_{\text{SiO}_2} x_{\text{CuO}_x} \quad (\text{Eq } 1)$$

$$x_{\text{CuO}_x}^{\text{corrected}} = x_{\text{CuO}_x} + 0.018 \cdot 4 \cdot x_{\text{SiO}_2} x_{\text{CuO}_x} \quad (\text{Eq } 2)$$

To establish the secondary standard correction in the whole PbO-CuO_x-SiO₂ system, it is also needed to take into account the contributions of the systems PbO-SiO₂^[11,12] (based on the stoichiometric compounds PbSiO₃, Pb₂SiO₄, and K-456 NIST glass standards):

$$x_{\text{SiO}_2}^{\text{corrected}} = x_{\text{SiO}_2} - x_{\text{SiO}_2} x_{\text{PbO}} \times \left(-0.0129 - 0.0206 x_{\text{SiO}_2} + 0.421 x_{\text{SiO}_2}^2 - 0.4066 x_{\text{SiO}_2}^3 \right), \quad (\text{Eq } 3)$$

and PbO-CuO_x (based on the Cu₂PbO₂ compound, assumed to be stoichiometric in the present study):

$$x_{\text{CuO}_{0.5}}^{\text{corrected}} = x_{\text{CuO}_{0.5}} - 0.0336 x_{\text{CuO}_{0.5}} x_{\text{PbO}}, \quad (\text{Eq } 4)$$

The total correction to improve the accuracy of phase compositions measured in the ternary system was introduced similarly to^[10] using the following expression for correction to project the binary expressions (1–4) into the ternary plane:

$$x_{\text{SiO}_2}^{\text{corrected}} = x_{\text{SiO}_2} - 0.018 \cdot 4 \cdot x_{\text{SiO}_2} x_{\text{CuO}_x} - x_{\text{SiO}_2} x_{\text{PbO}} \left(-0.0129 - 0.0206 x_{\text{SiO}_2} + 0.421 x_{\text{SiO}_2}^2 - 0.4066 x_{\text{SiO}_2}^3 \right), \quad (\text{Eq } 5)$$

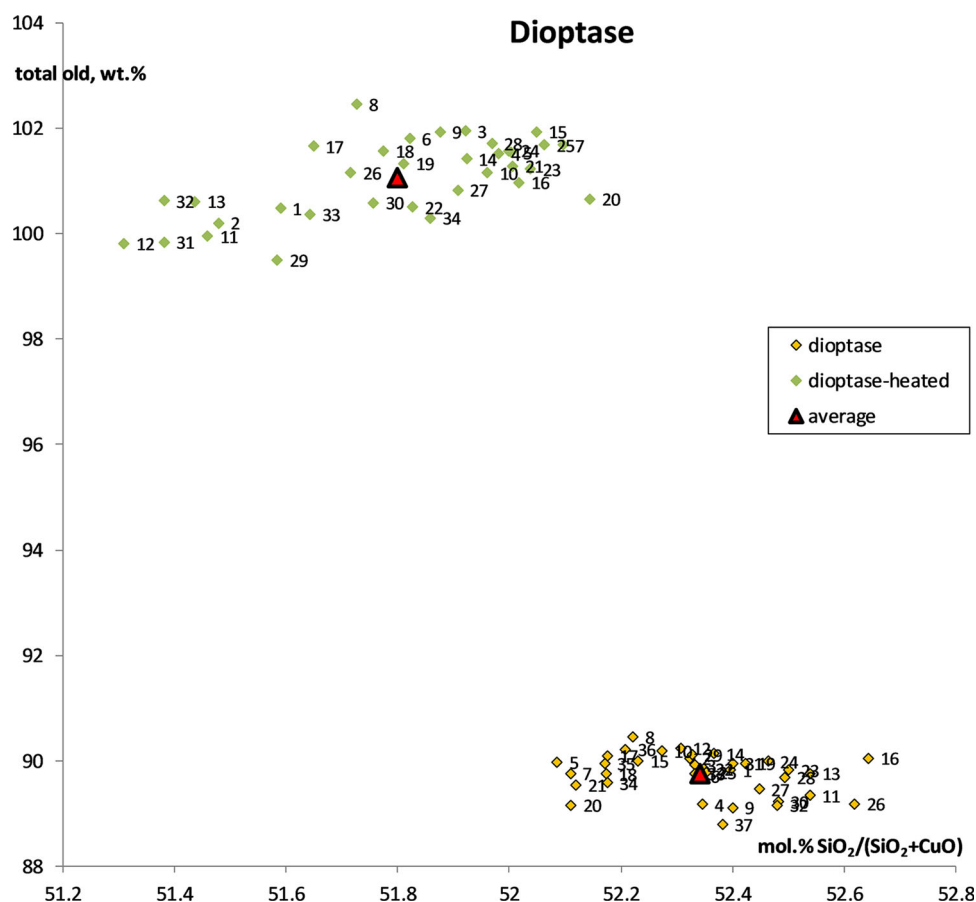
$$x_{\text{CuO}_{0.5}}^{\text{corrected}} = x_{\text{CuO}_{0.5}} \cdot \left(1 - x_{\text{SiO}_2}^{\text{corrected}} \right) / \left(1 - x_{\text{SiO}_2} \right) - 0.0336 x_{\text{CuO}_{0.5}} x_{\text{PbO}}, \quad (\text{Eq } 6)$$

$$x_{\text{PbO}}^{\text{corrected}} = 1 - x_{\text{SiO}_2}^{\text{corrected}} - x_{\text{CuO}_{0.5}}^{\text{corrected}} \quad (\text{Eq } 7)$$

where x_{SiO_2} , x_{PbO} , and $x_{\text{CuO}_{0.5}}$ are the initial molar fractions of SiO₂, PbO, and CuO_{0.5} with ZAF correction only. This correction is used in addition to the correction based on the dehydrated diopside standard described above.

A special experiment undertaken in the present study with the 50–100 micron size grains of quartz pressed together with Cu powder at room temperature (unreacted couple) identified that apparent 1.05 mol.% “CuO_{0.5}” is observed due to the secondary x-ray fluorescence in pure quartz grains surrounded by Cu. The following correction for the secondary x-ray fluorescence therefore has been developed and applied in the present study to the compositions of tridymite and cristobalite measured by EPMA:

Fig. 1 Results of EPMA analysis of the original diopside $\text{CuSiO}_3 \cdot \text{H}_2\text{O}$ and dehydrated at 650 °C diopside CuSiO_3



$$x(\text{CuO}_{0.5} \text{ in } \text{SiO}_2)^{\text{corr}} = x(\text{CuO}_{0.5} \text{ in } \text{SiO}_2)^{\text{initial}} - 0.0093 \text{ wt. fraction}(\text{CuO}_{0.5} \text{ in slag}) \quad (\text{Eq } 8)$$

Note that the Eq. (8) is not related to (1-7) as it only applies to very low-Cu phases, such as tridymite and cristobalite, surrounded by high-Cu slag (or metal).

2.2 Confirmation of Achievement of Equilibrium

To ensure the achievement of equilibrium in the samples, the “4-point test”^[4,17] was used including: (1) variation of equilibration time; (2) assessment of the compositional homogeneity of the phases by EPMA; (3) approaching the final equilibrium point from different starting conditions; and, importantly, (4) consideration of reactions specific to this system that may affect the achievement of equilibrium or reduce the accuracy. Several dedicated series of experiments were carried out for these purposes, resulted in the experimental design features described below.

Lead silicate master-slag (the eutectic at 40 mol.% PbO melting < 1013 K (740 °C)) was used to reduce Pb/PbO vaporization during the initial heating stage, to suppress foaming of the slag, and to enhance achievement of

homogeneity of the samples. Consequently, all initial mixtures were prepared from ($\text{SiO}_2 + \text{Cu}_2\text{O} + \text{master-slag} + \text{Cu} + \text{optional Pb}$) or ($\text{PbO} + \text{Cu}_2\text{O} + \text{master-slag} + \text{Cu} + \text{optional Pb}$) powders.

Preliminary experiments and analysis indicated that the major source of incomplete achievement of equilibrium in the present study was failure of some phases to nucleate from viscous liquid slag at low temperatures. The following procedure approaching equilibrium from low temperature was introduced to overcome this issue: several powder mixtures have been prepared for quartz (SiO_2), PbSiO_3 , and Pb_2SiO_4 primary phase fields, placed in Cu foil envelopes, sealed in vacuumed silica ampoules, pre-heated for a few minutes at their assumed liquidus temperatures 1003–1073 K (730–800 °C), cooled slightly below their assumed solidus 923–953 K (650–680 °C), and held up to 2–3 months. At the completion of the equilibration, the ampoules were slowly cooled, broken, and a part of each sample analyzed by SEM and EPMA for the absence of residual glass phase. In some cases, the glass was still present, so the samples were again wrapped in Cu foil, sealed in vacuumed ampoules, and kept at 923–953 K (650–680 °C) for several months, until the crystallization

Fig. 2 Backscattered scanning electron micrographs of quenched slag in the PbO-“CuO_{0.5}”-SiO₂ system in equilibrium with liquid or/and solid metallic Cu/Cu-Pb alloys.: (a) Liquid + Pb₇Si₆O₁₉ (non-equilibrium) + Cu, (b) Liquid + massicot (PbO) + Cu₂PbO₂ + Cu, (c) Liquid + alamosite PS (PbSiO₃) + cuprite (Cu₂O) + Cu, (d) PbO-“CuO_{0.5}”: Liquid + massicot (PbO) + Cu, (e) Liquid + alamosite PS (PbSiO₃) + P₂S (Pb₂SiO₄) + Cu, (f) Liquid + P₂S (Pb₂SiO₄) + Cu (s.) + Pb (liq.) and (g) Two liquids + cristobalite (SiO₂) + Cu

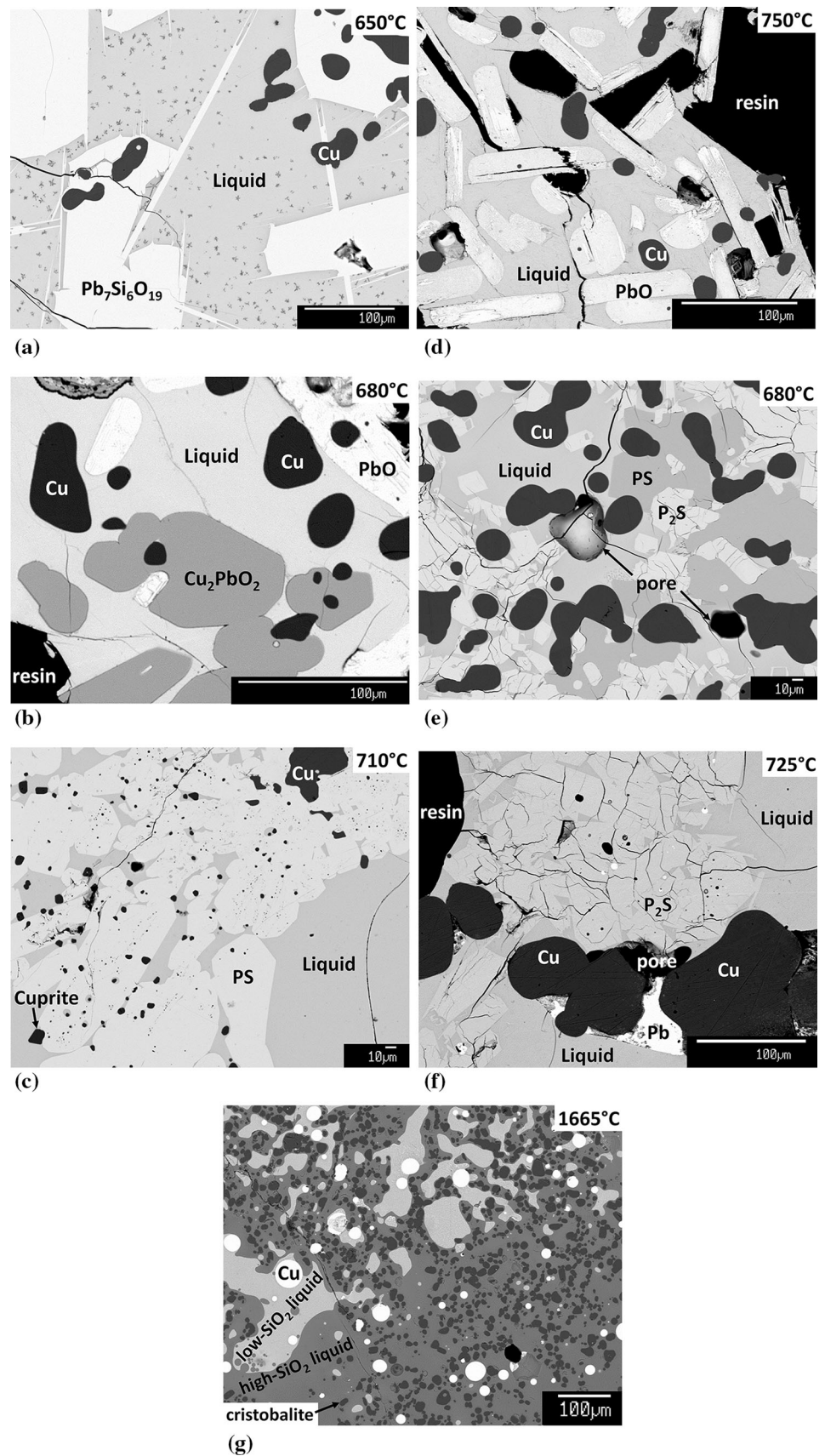


Table 1 Solid phases in the PbO–“CuO_{0.5}”–SiO₂ system

Phase name	Formula	Abbrev.	Public FactSage	Present study	Composition range, mol.%
Cristobalite	SiO ₂		Yes	Yes	< 0.45% “CuO _{0.5} ”, < 0.02% PbO(a)
Tridymite	SiO ₂		Yes	Yes	< 0.45% “CuO _{0.5} ”, < 0.02% PbO(a)
Quartz	SiO ₂		Yes	Yes	< 0.13% “CuO _{0.5} ”, < 0.02% PbO(a)
Cuprite	Cu ₂ O		Yes	Yes	< 0.04% SiO ₂ (a), < 0.05% PbO(a)
Tenorite	CuO		Yes	No	(b)
Barysilite	(9-x)PbO·xCuO _{1+y} ·6SiO ₂		No	Metast.	39.6–40.2% SiO ₂ , 0.65–4.1% “CuO _{0.5} ”
Copper plumbite	Cu ₂ PbO ₂		No	Yes	33.2–33.8% PbO < 0.01% SiO ₂ (a)
Lead metasilicate, alamosite	PbO·SiO ₂	PS	Yes	Yes	49.3–49.8% SiO ₂ < 0.4% “CuO _{0.5} ”
7:6 lead silicate	7PbO·6SiO ₂	P ₇ S ₆	No	Metast.	45.2–45.5% SiO ₂ < 0.3% “CuO _{0.5} ”
Lead orthosilicate	2PbO·SiO ₂	P ₂ S	Yes	Yes	32.7–33.5% SiO ₂ , < 0.7% “CuO _{0.5} ”
11:3 lead silicate	11PbO·3SiO ₂	P ₁₁ S ₃	Pb ₄ SiO ₆	Yes	~ 21.4% SiO ₂ < 0.3% “CuO _{0.5} ”
5-lead silicate	5PbO·SiO ₂	P ₅ S	No	No	(b)
Massicot	PbO		Yes	Yes	< 0.01% SiO ₂ (a), < 1.0% “CuO _{0.5} ”

All concentrations are in mol.%

(a) Within the uncertainty limits. (b) No data available

was complete. Finally, the samples were split into several smaller parts, and each part was equilibrated for 1–2 week at the temperature where liquid phase and solid phase(s) were estimated to coexist; preheating above the equilibration temperature was carefully avoided.

It should be noted that the Pb₇Si₆O₁₉ and barysilite Pb₈(Pb,Cu)Si₆O₂₁ phases were observed in several intermediate stages of the long heat treatments described above. However, they disappeared at final equilibration above solidus temperature. Sometimes, their residues (metastable according to the Gibbs phase rule) were observed in presence of liquid, PbSiO₃, Pb₂SiO₄, and Cu. So, both these phases are considered metastable at temperatures where liquid slag exists, > 943 K (670 °C). The prove that they become stable at lower temperatures is outside the scope of this study.

3 Results and Discussion

Examples of micrographs of quenched PbO–“CuO_{0.5}” and PbO–“CuO_{0.5}”–SiO₂ samples are given in Fig. 2(a), (b), (c), (d), (e), (f) and (g). All phases observed in the system and their measured compositions are listed in Table 1. Note that the EPMA results do not give accurate information on the oxygen concentration, so they are projected onto the PbO–“CuO_{0.5}”–SiO₂ plane.

The results of experiments, and measurements of phase compositions after correction according to Eq. 1–2, are reported in Table 2 (PbO–“CuO_{0.5}”–SiO₂) and Table 3 (PbO–“CuO_{0.5}”). The phase diagram for the pseudo-ternary PbO–“CuO_{0.5}”–SiO₂ based on the data from Shevchenko, Nicol et al.^[11] and present results is given in Fig. 3(a). The pseudo-binary diagram PbO–“CuO_{0.5}” is shown in Fig. 3(b). The estimated liquidus invariant points involving the observed phases are listed in Table 4.

The liquidus in quartz, tridymite and cristobalite (SiO₂) primary phase fields has been determined over a wide range of temperatures, 973–1913 K (700–1640 °C) and Cu/(Cu + Pb) molar ratios in slag (from 0 at the binary PbO–SiO₂ system^[11] to ~ 0.85). The quartz–tridymite and tridymite–cristobalite transitions were not determined directly and accepted from FactSage database^[23] as 1144 K (867 °C) and 1742 K (1465 °C), respectively. A range of liquid immiscibility is known to exist in the “CuO_{0.5}”–SiO₂ system and this is expected to extend over a part of the ternary PbO–“CuO_{0.5}”–SiO₂ system but not to reach the binary PbO–SiO₂.^[11] The phase equilibria in this range of compositions are extremely difficult to characterize. Only one sample at 1938 K (1665 °C) (see Fig. 3a) has reached the two immiscible liquids range. This means that the area of two immiscible liquids does not extend to more than 6 mol.% PbO, which is much less than predicted by a

Table 2 Experimental compositions of phases in the PbO–⁷³SiO₂ system in equilibrium with liquid Cu–Pb or/and solid Cu metallic phase

Exp. no.	Substrate	Premelt/precrystallization T, K, °C	Final equilibration T, K, °C	Time, h(a)	Phase	Composition, mol. %		
						SiO ₂	PbO (Pb)	CuO _{0.5} (Cu)
<i>SiO₂ (quartz and tridymite) liquidus</i>								
1	SiO ₂ amp.	No	1023 (750)	120	Liquid	57.6 ± 0.17	40.4 ± 0.18	2.0 ± 0.16
					Quartz	99.85 ± 0.06	0.02 ± 0.02	0.13 ± 0.06
					Cu metal	n.a.	n.a.	~100
2	SiO ₂ amp.	No	1073 (800)	120	Liquid	57.5 ± 0.18	39.5 ± 0.19	3.0 ± 0.15
					Quartz	99.89 ± 0.08	0.01 ± 0.01	0.10 ± 0.08
					Cu metal	n.a.	n.a.	~100
3	SiO ₂ amp. + hole	1473 (1200); 1373 (1100)	1423 (1150)	0.2 + 1+3	Liquid	19.9 ± 0.19	5.6 ± 0.10	74.5 ± 0.2
					Tridymite	99.94	0.00	0.06
					Pb-Cu metal	0	0	100.0
4	SiO ₂ amp.	1603 (1330)	1573 (1300)	0.5 + 3	Liquid	54.8 ± 0.3	25.6 ± 0.2	19.7 ± 0.3
					Tridymite	99.77 ± 0.05	0.02 ± 0.02	0.22 ± 0.06
					Pb-Cu metal	0.1 ± 0.02	0.1 ± 0.02	99.9 ± 0.02
5	SiO ₂ open	1603 (1330)	1573 (1300)	0.25 + 1.5	Liquid	59.6 ± 0.2	27.0 ± 0.2	13.4 ± 0.15
					Tridymite	99.75 ± 0.13	0.01 ± 0.01	0.24 ± 0.13
					Pb-Cu metal	0.6	0.6	99.4
6	SiO ₂ open	1603 (1330)	1573 (1300)	0.25 + 1.5	Liquid	59.7 ± 0.2	27.7 ± 0.19	12.6 ± 0.15
					Tridymite	99.75 ± 0.13	0.01 ± 0.01	0.24 ± 0.13
					Pb-Cu metal	0.6	0.6	99.4
7	SiO ₂ open	1603 (1330)	1573 (1300)	0.25 + 1.5	Liquid	50.2 ± 0.2	21.9 ± 0.3	27.9 ± 0.3
					Tridymite	99.75 ± 0.03	0.01 ± 0.01	0.24 ± 0.02
					Pb-Cu metal	n.a.	n.a.	59.7 ± 0.9
8	SiO ₂ open	1603 (1330)	1573 (1300)	0.25 + 1.5	Liquid	30.4 ± 0.8	9.8 ± 0.2	59.7 ± 0.9
					Tridymite	n.a.	n.a.	99.96 ± 0.03
					Pb-Cu metal	0.04 ± 0.03	0.04 ± 0.03	10.1 ± 0.3
9	SiO ₂ open	1693 (1420)	1673 (1400)	0.25 + 1.5	Liquid	64.8 ± 0.2	25.1 ± 0.3	10.1 ± 0.3
					Tridymite	99.86 ± 0.05	0.00	0.14 ± 0.05
					Pb-Cu metal	1.1 ± 0.3	1.1 ± 0.3	98.9 ± 0.3
10	SiO ₂ open	1693 (1420)	1673 (1400)	0.25 + 1.5	Liquid	63.1 ± 0.5	23.2 ± 0.5	13.7 ± 0.7
					Tridymite	99.72 ± 0.10	0.01 ± 0.01	0.26 ± 0.09
					Pb-Cu metal	0.08 ± 0.02	0.08 ± 0.02	99.92 ± 0.02
11	SiO ₂ open	1693 (1420)	1673 (1400)	0.25 + 1.5	Liquid	34.9 ± 0.4	9.3 ± 0.4	55.8 ± 0.6
					Tridymite	99.54 ± 0.07	0.01 ± 0.01	0.45 ± 0.07
					Pb-Cu metal	0.0	0.0	100.0
<i>SiO₂ (cristobalite) liquidus</i>								
12	SiO ₂ amp. + hole	No	1743 (1470)	2	Liquid	67.7 ± 0.2	22.4 ± 0.2	9.9 ± 0.2
					Cristobalite	99.84 ± 0.03	0.01 ± 0.01	0.14 ± 0.02

Table 2 continued

Exp. no.	Substrate	Premelt/precrystallization T, K, °C	Final equilibration T, K, °C	Time, h(a)	Phase	Composition, mol. %		
						SiO ₂	PbO (Pb)	CuO _{0.5} (Cu)
13	SiO ₂ amp. + hole	No	1748 (1475)	2	Pb-Cu metal Liquid Cristobalite	39.4 ± 0.4 99.55 ± 0.02	0.8 9.5 ± 0.3 0.00	99.2 51.1 ± 0.5 0.45 ± 0.02
14	SiO ₂ amp. + hole	No	1773 (1500)	2	Pb-Cu metal Liquid Cristobalite	62.1 ± 0.11 99.81 ± 0.02	19.3 ± 0.12 0.02 ± 0.01	99.99 18.6 ± 0.18 0.17 ± 0.02
15	SiO ₂ amp.	No	1873 (1600)	1	Pb-Cu metal Liquid Cristobalite	58.2 ± 0.3 99.61	n.a. 10.8 ± 0.16 0.02	~99 31.0 ± 0.3 0.37
16	SiO ₂ amp.	1903 (1630)	1898 (1625)		Pb-Cu metal Liquid Cristobalite	76.9 ± 0.2 99.80	0.02 10.1 ± 0.3 0.00	99.98 13.0 ± 0.3 0.20
17	SiO ₂ amp.	1918 (1645)	1913 (1640)		Pb-Cu metal Liquid Cristobalite	82.2 ± 0.6 99.73 ± 0.07	0.06 ± 0.02 6.6 ± 0.2 0.01 ± 0.01	99.94 ± 0.02 11.2 ± 0.4 0.26 ± 0.08
18	SiO ₂ amp.	No	1933 (1660)	0.7	Pb-Cu metal Liquid Cristobalite	57.8 ± 0.7 n.a.	0.06 ± 0.05 6.9 ± 0.11	99.94 ± 0.05 35.3 ± 0.7
19	SiO ₂ amp.	1943 (1670)	1938 (1665)	0.1 + 0.7	Pb-Cu metal Liquid1 Liquid2 Cristobalite Pb-Cu metal	86.3 ± 0.6 61.3 ± 0.3 99.57	0.01 2.4 ± 0.09 5.6 ± 0.05 0.01 0.03	99.99 11.3 ± 0.6 33.2 ± 0.3 0.42 99.97
<i>Tridymite (SiO₂) + cuprite (Cu₂O) boundary</i>								
20	SiO ₂ open	1423 (1150)	1323 (1050)	0.25 + 2	Liquid Tridymite Cuprite	36.8 ± 0.3 99.75 0.0	19.9 ± 0.2 0.00 0.0	43.3 ± 0.3 0.25 100.0
21	SiO ₂ open	1423 (1150)	1373 (1100)	0.25 + 1.5	Liquid Tridymite Cuprite	30.7 ± 0.2 99.60 ± 0.08 0.00	14.2 ± 0.2 0.03 0.02	55.1 ± 0.2 0.37 ± 0.08 99.98
22	SiO ₂ amp + hole	1473 (1200) + 1393 (1120)	1418 (1145)	0.1 + 0.3 + 3	Pb-Cu metal Liquid Tridymite Cuprite Pb-Cu metal	24.9 ± 0.4 99.84 ± 0.2 0.04 ± 0.02	8.7 ± 0.3 0.00 0.02 ± 0.01 0.05	99.95 66.4 ± 0.5 0.16 ± 0.2 99.92 ± 0.01 99.95
<i>Cuprite (Cu₂O) liquidus</i>								
23	Cu net	No	973 (700)	168	Liquid Cuprite	37.7 ± 0.3 0.01 ± 0.01	51.8 ± 0.4 0.01 ± 0.01	10.5 ± 0.4 99.98 ± 0.01

Table 2 continued

Exp. no.	Substrate	Premelt/precrystallization T, K, °C	Final equilibration T, K, °C	Time, h(a)	Phase	Composition, mol. %		
						SiO ₂	PbO (Pb)	CuO _{0.5} (Cu)
24	Cu net	1023 (750)	973 (700)	2 + 48	Liquid	22.8 ± 0.3	59.8 ± 0.4	17.4 ± 0.5
					Cuprite	0.00	0.02 ± 0.02	99.98 ± 0.02
25	Cu net	1073 (800)	973 (700)	1 + 48	Liquid	6.4 ± 0.12	64.0 ± 0.4	29.6 ± 0.5
					Cuprite	0.0	0.0	100.0
26	Cu net	1073 (800)	973 (700)	1 + 48	Liquid	6.9 ± 0.3	65.1 ± 0.3	28.0 ± 0.3
					Cuprite	0.0	0.0	100.0
27	Cu net	1023 (750)	973 (700)	2 + 48	Liquid	38.2 ± 0.3	51.4 ± 0.4	10.4 ± 0.4
					Cuprite	0.0	0.0	100.0
28	Cu net	1073 (800)	1023 (750)	1 + 24	Liquid	1.3 ± 0.3	52.7 ± 1.9	46.0 ± 1.8
					Cuprite	0.0	0.0	100.0
29	Cu net	1083 (810)	1023 (750)	1 + 18	Liquid	5.9 ± 0.12	59.0 ± 0.6	35.1 ± 0.6
					Cuprite	0.02	0.02	99.96
30	Cu net	1073 (800)	1023 (750)	1 + 48	Liquid	36.5 ± 0.3	49.9 ± 0.3	13.6 ± 0.3
					Cuprite	0.01 ± 0.01	0.03 ± 0.01	99.96 ± 0.01
31	Cu net	1103 (830)	1073 (800)	1 + 18	Liquid	20.9 ± 0.18	53.5 ± 0.4	25.6 ± 0.4
					Cuprite	0.00	0.04 ± 0.02	99.96 ± 0.03
32	Cu net	1273 (1000)	1223 (950)	0.5 + 3	Liquid	2.8 ± 0.14	36.9 ± 0.7	60.3 ± 0.7
					Cuprite	0.00	0.03 ± 0.02	99.97 ± 0.02
<i>Cuprite (Cu₂O) + alamosite PS (PbSiO₃) boundary</i>								
33	Cu foil in SiO ₂ amp.	No	953 (680)	168	Liquid	36.1 ± 0.6	54.5 ± 0.5	9.4 ± 0.9
					Cuprite	0.00	0.02	99.98
34	Cu foil in SiO ₂ amp.	No	973 (700)	168	Alamosite	49.5 ± 0.18	50.2 ± 0.2	0.3 ± 0.3
					Liquid	53.5 ± 0.2	38.7 ± 0.15	7.7 ± 0.2
					Cuprite	0.04 ± 0.02	0.03 ± 0.01	99.94 ± 0.03
35	Cu foil in SiO ₂ amp.	No	983 (710)	168	Alamosite	49.4 ± 0.4	50.2 ± 0.2	0.4 ± 0.3
					Liquid	50.5 ± 0.2	40.8 ± 0.19	8.7 ± 0.2
					Cuprite	0.03	0.05	99.92
					Alamosite	49.4 ± 0.3	50.3 ± 0.3	0.3 ± 0.3
<i>Cuprite (Cu₂O) + alamosite PS (PbSiO₃) + P₂S (Pb₂SiO₄) eutectic</i>								
36	Cu foil in SiO ₂ amp.	No	943 (670)	336	Liquid	34.7 ± 0.3	55.3 ± 0.5	10.0 ± 0.6
					Cuprite	0.01 ± 0.01	0.02 ± 0.02	99.97 ± 0.02
					Alamosite	49.3 ± 0.2	50.3 ± 0.3	0.37 ± 0.4
					Pb ₂ SiO ₄	33.0 ± 0.4	66.5 ± 0.6	0.45 ± 0.4
					Barysilite(b)	40.2 ± 0.10	55.7 ± 0.2	4.1 ± 0.3
<i>Alamosite PS (PbSiO₃) + P₂S (Pb₂SiO₄) boundary</i>								
37	Cu foil in SiO ₂ amp.	No	953 (680)	168	Liquid	35.4 ± 0.3	53.5 ± 0.18	11.0 ± 0.4
					Alamosite	49.7 ± 0.2	50.1 ± 0.2	0.21 ± 0.19
					Pb ₂ SiO ₄	33.5 ± 0.2	66.3 ± 0.2	0.21 ± 0.2

Table 2 continued

Exp. no.	Substrate	Premelt/precrystallization T, K, °C	Final equilibration T, K, °C	Time, h(a)	Phase	Composition, mol. %		
						SiO ₂	PbO (Pb)	CuO _{0.5} (Cu)
38	Cu foil in SiO ₂ amp.	No	973 (700)	168	Liquid	38.2 ± 0.2	56.3 ± 0.12	5.5 ± 0.2
<i>Cuprite + Pb₂SiO₄ boundary</i>	Cu net	988 (715)	933 (660)	2 + 72	Alamosite	49.8 ± 0.2	50.1 ± 0.16	0.07 ± 0.18
					Pb ₂ SiO ₄	32.7 ± 0.4	67.2 ± 0.2	0.12 ± 0.17
39	Cu net	988 (715)	933 (660)	2 + 72	Liquid	21.0 ± 0.3	63.2 ± 0.5	15.8 ± 0.5
<i>Pb₂SiO₄ liquidus</i>	Ir wire	988 (715)	953 (680)	1 + 22	Cuprite	0.0	0.0	100.0
					Pb ₂ SiO ₄	33.3 ± 0.3	66.0 ± 0.3	0.72 ± 0.3
40	Ir wire	988 (715)	953 (680)	1 + 22	Liquid	20.9 ± 0.13	66.2 ± 0.2	12.9 ± 0.2
<i>Pb-Cu (liq) + Cu (sol) + P₂S (Pb₂SiO₄) boundary</i>	Cu foil in SiO ₂ amp.	No	988 (715)	168	Pb ₂ SiO ₄	33.1 ± 0.2	66.5 ± 0.2	0.33 ± 0.14
					Liquid	40.4 ± 0.19	58.1 ± 0.2	1.5 ± 0.17
41					Pb ₂ SiO ₄	33.2 ± 0.2	66.6 ± 0.2	0.19 ± 0.16
					Barysilite(b)	39.6 ± 0.06	59.8 ± 0.14	0.65 ± 0.09
42					Cu solid		0.03 ± 0.01	99.97 ± 0.01
					Pb-Cu liquid		99.65 ± 0.2	0.35 ± 0.2
43					Liquid	39.4 ± 0.2	58.8 ± 0.2	1.8 ± 0.11
					Pb ₂ SiO ₄	33.1 ± 0.3	66.8 ± 0.2	0.09 ± 0.15
<i>Pb₂SiO₄ + Pb₁₁Si₃O₁₇ + Cu₂PbO₂ eutectic</i>					Cu solid		0.05 ± 0.03	99.95 ± 0.03
					Pb-Cu liquid		99.5 ± 0.4	0.5 ± 0.4
44	Cu foil	No	933 (660)	3	Liquid	20.0 ± 0.14	66.2 ± 0.17	13.8 ± 0.13
<i>PbO (massicot) liquidus</i>	Cu net	988 (715)	953 (680)	2 + 72	Pb ₂ SiO ₄	33.4 ± 0.2	66.6 ± 0.3	0.05 ± 0.05
					Pb ₁₁ Si ₃ O ₁₇	21.4 ± 0.08	78.4 ± 0.18	0.27 ± 0.12
45	Ir wire	993 (720)	973 (700)	1 + 22	Cu ₂ PbO ₂	0.00	33.8 ± 0.3	66.2 ± 0.3
46	Ir wire	1073 (800)	1023 (750)	0.5 + 22	Liquid	13.3 ± 0.17	69.4 ± 0.2	17.3 ± 0.3
					Massicot	0.00	99.23 ± 0.2	0.77 ± 0.2
47	Ir plate	1073 (800)	1023 (750)	1 + 24	Liquid	9.6 ± 0.2	66.1 ± 0.4	24.3 ± 0.5
					Massicot	0.00	99.79 ± 0.05	0.21 ± 0.05
<i>PbO (massicot) + Cu₂PbO₂ boundary</i>	Cu net	988 (715)	953 (680)	2 + 72	Liquid	3.7 ± 0.12	69.4 ± 0.2	26.9 ± 0.2
					Massicot	0.00	99.77	0.23
48	Cu net	988 (715)	953 (680)	2 + 72	Liquid	1.8 ± 0.13	67.8 ± 0.4	30.4 ± 0.4
					Massicot	0.00	99.56 ± 0.09	0.44 ± 0.09
					Cu ₂ PbO ₂	0.00	33.2 ± 0.09	66.8 ± 0.09

Table 2 continued

Exp. no.	Substrate	Premelt/precrystallization T, K, °C	Final equilibration T, K, °C	Time, h(a)	Phase	Composition, mol. %		
						SiO ₂	PbO (Pb)	CuO _{0.5} (Cu)
49	Ir wire	993 (720)	953 (680)	1 + 22	Liquid	14.4 ± 0.18	68.4 ± 0.3	17.2 ± 0.3
					Massicot	0.00	99.39 ± 0.03	0.61 ± 0.03
					Cu ₂ PbO ₂	0.00	33.6 ± 0.07	66.4 ± 0.07

Italic values are those below the estimated noise level (0.1 mol. %)

amp. ampoule, *n.a.* not analyzed (due to small crystal size)

(a) Times are listed as (premelt time) + (equilibration time) or (precrystallization time) + (equilibration time). (b) Barysilite is considered metastable phase at the studied range of conditions

preliminary FactSage model using an internal PyroSearch database.^[24] A possible reason for the difference between the observations and predictions lies in the inaccuracy of properties of pure SiO₂ in the database.

The low-temperature part of cuprite (Cu₂O) primary phase field characterized in the present investigation contains the alamosite (PbSiO₃ = PS), Pb₂SiO₄ = P₂S, Pb₁₁-Si₃O₁₇ = P₁₁S₃, Cu₂PbO₂, and massicot (PbO) phase fields, with the following invariant points:

Liquid → Cuprite(Cu₂O) + Quartz(SiO₂)

+ Alamosite(PbSiO₃) at 973 ± 5 K (700 ± 5 °C),

Liquid → Cuprite(Cu₂O) + Alamosite(PbSiO₃)

+ Pb₂SiO₄ at 943 ± 5 K (670 ± 5 °C),

Liquid → Cuprite (Cu₂O) + Pb₂SiO₄

+ Pb₁₁Si₃O₁₇ at 930 ± 5 K (657 ± 5 °C),

Liquid + Cu₂PbO₂ → Cuprite(Cu₂O)

+ Pb₁₁Si₃O₁₇ at 933 ± 5 K (660 ± 5 °C),

(Liquid) + Massicot(PbO) + Cuprite (Cu₂O)

→ Cu₂PbO₂ at 973 ± 10 K (700 ± 10 °C).

It also has two saddle (maximum) points on the boundary lines with PbSiO₃ and Pb₂SiO₄:

Liquid → Cuprite(Cu₂O) + Alamosite (PbSiO₃) at 993

± 10 K (720 ± 10 °C),

Liquid → Cuprite(Cu₂O) + Pb₂SiO₄ at 958

± 5 K (685 ± 5 °C).

Cristobalite, tridymite and quartz were found to have limited solubility of copper oxide (Fig. 4), which reached 0.45 mol. % “CuO_{0.5}” for equilibrium with the slags of the highest copper concentration close to the Liquid + tridymite + cuprite boundary. None of the silica polymorphs (including the points in the PbO-SiO₂ system^[11]) were found to have any significant solubility of PbO; it was measured to be less than 0.03 mol. %, which is below the estimated minimum detection limit. As can be seen from comparison in Fig. 4, the differences of solubility of “CuO_{0.5}” in all silica polymorphs, as well as the differences between tridymite and cristobalite and the effects of temperature, are within the accuracy of the present measurements. The principal parameter that determines the solubility of “CuO_{0.5}” in silica polymorphs is the Cu/(Cu + Pb) molar ratio in slag.

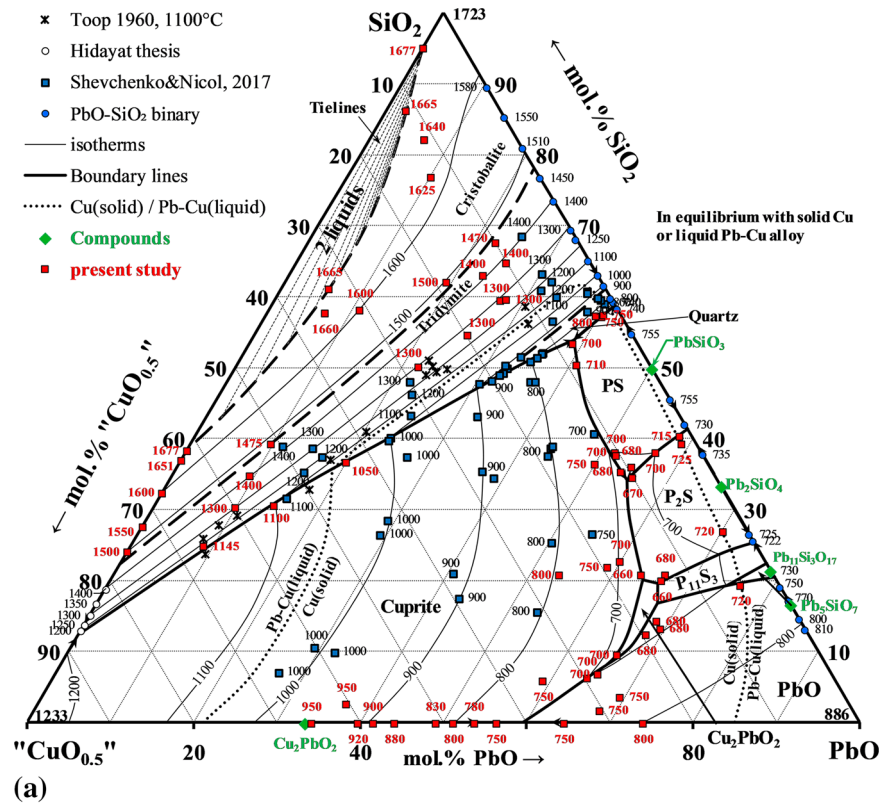
In summary, present experimental study has produced a comprehensive dataset for the Pb-Cu-Si-O system in equilibrium with metallic alloy that is essential for the lead and copper smelting and recycling industry, and provides valuable new data that can be used to optimize the thermodynamic databases of these systems.

Table 3 Experimental compositions of phases in the PbO–CuO_{0.5} system in equilibrium with solid Cu metallic phase

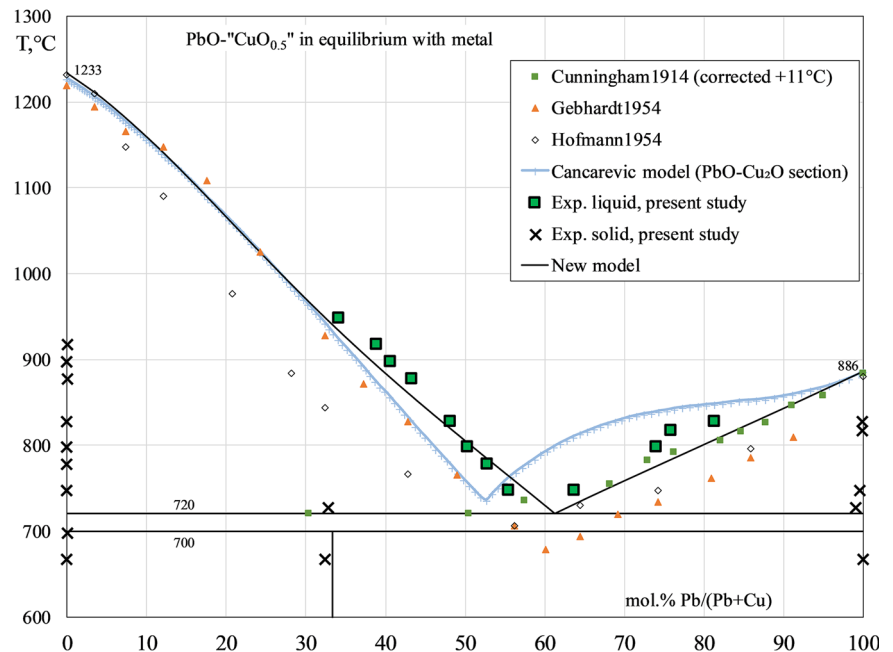
Exp. no.	Substrate	Premelt/precrySTALLIZATION T, K (°C)	Final equilibration T, K (°C)	Time, h(a)	Phase	Composition, mol.%	
						PbO (Pb)	CuO _{0.5} (Cu)
<i>Cuprite (Cu₂O) liquidus</i>							
1	Cu net	1073 (800)	1023 (750)	0.5 + 22	Liquid	56.3	43.7
					Cuprite	0.02	99.98
					Cu metal	0.00	100.00
2	Cu net	1073 (800)	1053 (780)	0.5 + 5	Liquid	53.7	46.3
					Cuprite	0.03	99.97
					Cu metal	0.01	99.99
3	Cu net	1103 (830)	1073 (800)	0.25 + 18	Liquid	51.2	48.8
					Cuprite	0.01	99.99
					Cu metal	0.01	99.99
4	Cu net	1123 (850)	1103 (830)	0.5 + 6	Liquid	49.0	51.0
					Cuprite	0.01	99.99
					Cu metal	0.01	99.99
5	Cu foil	1193 (920)	1153 (880)	0.25 + 4	Liquid	44.1	55.9
					Cuprite	0.05	99.95
					Cu metal	0.05	99.95
6	Cu foil	1203 (930)	1173 (900)	0.5 + 5	Liquid	41.5	58.5
					Cuprite	0.02	99.98
					Cu metal	0.03	99.97
7	Cu foil	1213 (940)	1193 (920)	0.25 + 3	Liquid	39.7	60.3
					Cuprite	0.02	99.98
					Cu metal	0.01	99.99
8	Cu net	1253 (980)	1223 (950)	1 + 2	Liquid	34.1	65.9
					Cuprite	n.a.	
					Cu metal	0.00	100.00
<i>Massicot (PbO) liquidus</i>							
9	Cu foil	1063 (790)	1013 (740)	0.25 + 4	Liquid	64.7	35.3
					Massicot	99.6	0.4
					Cu metal	0.04	99.96
10	Cu foil	1048 (775)	1023 (750)	0.2 + 4	Liquid	64.5	35.5
					Massicot	99.6	0.4
					Cu metal	0.04	99.96
11	Cu foil	1083 (810)	1073 (800)	0.25 + 3	Liquid	74.0	26.0
					Massicot	99.0	1.0
					Cu metal	0.11	99.89
12	Ir wire	1118 (845)	1103 (830)	0.25 + 3	Liquid	81.9	18.1
					Massicot	99.9	0.1
					Pb metal (liq)	83.3	16.7
					Cu metal	0.46	99.54
<i>Subsolidus</i>							
13	Cu foil	1173 (900)	1003 (730)	3 + 16	Cu ₂ PbO ₂	33.6	66.4
					Massicot	99.15	0.85
					Cu metal	0.01	99.99

Abbreviations same as in Table 2

Fig. 3 Liquidus surface of the pseudo-ternary PbO -“ $\text{CuO}_{0.5}$ ”- SiO_2 (a) and pseudo-binary PbO -“ $\text{CuO}_{0.5}$ ” (b) systems in equilibrium with metallic solid Cu or liquid Cu-Pb alloy, compared to results.^[1,18–22] All temperatures are in °C; red labels indicate present study, black—literature data. PbO - SiO_2 binary is based on Ref 11 (Color figure online)



(a)



(b)

Table 4 Invariant points of the PbO–“CuO_{0.5}”–SiO₂ liquidus in equilibrium with metal

Reaction	Estimated T, K, °C	Liquid composition, mol.%		
		PbO	“CuO _{0.5} ”	SiO ₂
Liquid → quartz + cuprite + alamosite/Cu(solid)	973(700) ± 10	39	7	54
Liquid + Pb–Cu(liquid) → quartz + alamosite + Cu(solid)	1003(730) ± 10	40	1	59
Liquid → cuprite + alamosite + P ₂ S/Cu(solid)	943(670) ± 5	55	11	34
Liquid + Pb–Cu(liquid) → alamosite + P ₂ S + Cu(solid)	985(712) ± 5	58	2	40
Liquid + cuprite → P ₂ S + Cu ₂ PbO ₂ /Cu(solid)	933(660) ± 10	64	16	20
Liquid + Pb–Cu(liquid) → P ₂ S + P ₁₁ S ₃ + Cu(solid)	983(710) ± 10	73	3	24
Liquid → P ₂ S + P ₁₁ S ₃ + Cu ₂ PbO ₂ /Cu(solid)	933(660) ± 10	66	14	20
Liquid + PbO → Cu ₂ PbO ₂ + P ₁₁ S ₃ /Cu(solid)	943(670) ± 10	67	16	17
Liquid + P ₅ S + Pb–Cu(liquid) → P ₁₁ S ₃ + Cu(solid)	983(710) ± 5	75	5	20
Liquid + massicot + Pb–Cu(liquid) → P ₅ S + Cu(solid)	988(715) ± 5	76	5	19
(Liquid) + massicot + cuprite → Cu ₂ PbO ₂ /Cu(solid)	973(700) ± 10	66	26	8

Cu(solid) means that the point is in equilibrium with solid Cu metal, but the amount of the latter does not change during it

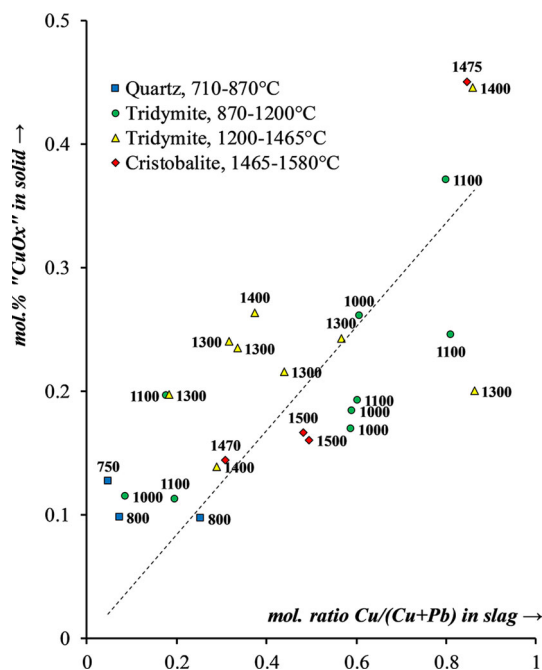


Fig. 4 Copper oxide concentrations in solid quartz, tridymite and cristobalite crystals in equilibrium with slags of the PbO–“CuO_{0.5}”–SiO₂ system and Cu–Pb metallic alloy, as a function of molar ratio Cu/(Cu + Pb) in the slag, including points from Ref 1, 18–22. Labels indicate the experimental temperatures (°C)

4 Conclusions

New phase equilibria information in the PbO–“CuO_{0.5}”–SiO₂ system in equilibrium with metal (solid Cu, liquid Pb–Cu alloy, or both) has been obtained. The investigation includes temperatures in the range 933–1938 K (660–1665 °C), and the equilibria of liquid slag with one or two

crystalline phases: quartz, tridymite, cristobalite (SiO₂), cuprite (Cu₂O), lead silicates (PbSiO₃, Pb₂SiO₄, Pb₁₁Si₃O₁₇); (d) lead oxide (massicot PbO); (e) copper plumbite (Cu₂PbO₂). This is the first systematic study of the Pb–Cu–Si–O system in equilibrium with metal, as a part of a broader study of the multicomponent system Pb–Zn–Fe–Cu–Si–Ca–Al–Mg–O for the lead smelting and recycling industries.

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